Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Levulinates from Lignocellulose as Bio-based Additives for Plastics (LevuLig)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), **Advanced Bio (X)**, Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals			
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	*Additives*		
Concept of Proposed Technology	*The proposed technology focuses on converting renewable hexoses into levulinic acid esters (levulinates) through acid-catalyzed reactions in alcoholic media, enabling high selectivity and minimal by-products. These levulinates serve as sustainable, biobased plasticizers for polymers like PVC, elastomers, and PLA, replacing harmful phthalates. Potential uses include improving flexibility and durability in plastics while supporting eco-friendly material development.*			
Rationale for Joint R&D Project	*Joint R&D is needed to test the performance of these biobased additives at laboratory scale, evaluating their plasticizing properties, migration behavior, and compatibility with different polymers. Additionally, collaboration ensures expertise in synthesis, toxicological assessment, and application testing, which is critical for meeting regulatory requirements and preparing for industrial-scale implementation.*			
Key Words	*1. Levulinates* *2. Bio-based Plasticizers* *3. Sustainable Polymers*			

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	1. *Develop a robust synthesis pathway starting from cellulose, utilizing alcohols as co-solvents to efficiently produce levulinic acid esters.* 2. *Design and optimize downstream processing for effective separation and purification of levulinates, ensuring high yield and purity.* 3. *Integrate levulinates into polymer formulations and systematically evaluate their performance as plasticizers, focusing on mechanical properties, migration behavior, and long-term stability.*			
R&D Target Specifications	• ***Levulinate Yield and Purity**: Achieve a conversion yield of ≥70% from cellulose and a product purity of >95% after downstream processing.* • ***Application Testing**: Validate performance in PVC, PLA, and elastomer formulations at concentrations of 10%, 20%, and 30%, maintaining mechanical integrity and thermal stability.*			
R&D Plan	***Phase I – Pre-study (6 months)*** ***Objective:** Establish feasibility and define process parameters.* ***Key Tasks & Methodologies:*** • *Literature review and benchmarking of cellulose-to-levulinate conversion routes.* • *Lab-scale experiments using alcohols as co-solvents to identify optimal reaction conditions.* • *Preliminary analysis of potential downstream separation techniques (filtration, extraction).* ***Expected Milestones:*** • *Selection of most promising synthesis pathway.* • *Initial yield and purity data for levulinates.* • *Draft process concept for scale-up* *Phase II – Main R&D (30 months)* ***Objective:** Develop and validate scalable processes and product applications.* ***Key Tasks & Methodologies:*** • ***Months 7–18:** Optimize synthesis parameters for high yield and selectivity; design and test pilot-scale reaction system.* • ***Months 19–24:** Develop and implement downstream purification achieving >95% purity* • ***Months 25–30:** Formulate levulinates into PVC, PLA, and elastomer compounds; perform mechanical, migration, and aging tests.(partner needed)* ***Expected Milestones:*** • *Demonstration of synthesis process with ≥70% yield.* • *Validated downstream process* • *Performance data confirming levulinates as effective plasticizers in target polymers.*			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	***Research Institute (CBP)*** ***Technical Contributions:*** • *Develop and optimize the synthesis pathway for levulinic acid esters starting from cellulose using alcohols as co-solvents.* • *Conduct lab-scale and pilot-scale experiments to establish reaction conditions and validate process feasibility.* ***Research Scope:*** • *Focus on chemical conversion, reaction engineering, and preliminary downstream separation concepts.* ***Resources Provided:*** • *Laboratory infrastructure, analytical equipment, and expertise in biomass conversion and process development.* ***Specific Tasks:*** • *Design and execute synthesis trials.* • *Generate initial yield and purity data.* • *Support scale-up concepts for continuous processing.*
Expected R&R of Korean SME Partner	***Korean SME*** ***Polymer Application Partner:*** • *Formulate levulinates into PVC, PLA, and elastomer compounds and perform mechanical, migration, and aging tests.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	***Technological Achievements:*** *Development of a robust synthesis process for levulinic acid esters from cellulose, optimized downstream purification achieving >95% purity, and validated application of levulinates as effective plasticizers in PVC, PLA, and elastomers.* ***Commercialization Potential:*** *Creation of sustainable, biobased alternatives to phthalates with strong market demand in the plastics industry, enabling compliance with tightening environmental and health regulations.* ***Broader Industrial and Economic Impact:*** *Strengthening the bioeconomy by utilizing renewable resources, reducing dependency on fossil-based additives, and opening new value chains for green chemicals and materials. This supports circular economy goals and enhances competitiveness in polymer markets.*

6. References

(1) Köchermann, J.; Klüpfel, C.; Klemm, M.: Brønsted/lewis-acid combinations for hydrothermal production of levulinic acid from starch residues. European Biomass Conference and Exhibition Proceedings, 2020, S. 515-519.

(2) Köchermann, J.; Schreiber, J.; Klemm, M.: Conversion of d-Xylose and Hemicellulose in Water/Ethanol Mixtures. ACS Sustainable Chemistry & Engineering 7 (2019) H. 14, S. 12323-12330.

(3) Körner, P.; Köchermann, J.: Production of levulinates from starch-containing residues. 3th International Bioeconomy Congress, Stuttgart 21./22.09.2020.

(4) Köchermann, J.; Mühlenberg, J.; Klemm, M.: Kinetics of Hydrothermal Furfural Production from Organosolv Hemicellulose and D-Xylose. Industrial & Engineering Chemistry Research 57 (2018) H. 43, S. 14417-14427.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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